Real Determinacy Screens for DSGE Estimation

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Abstract

Bayesian DSGE estimation imposes determinacy as a support restriction on the posterior. This paper shows that a common numerical rule, rejecting draws whose complex-QZ extraction returns a complex policy matrix, is not a valid support restriction. For a real linear rational-expectations pencil, complex stable roots occur in conjugate pairs and span a real stable deflating subspace. If the stable-root count and graph condition hold, the DSGE policy rule is real; non-zero imaginary entries in a complex extraction reflect basis recovery error, not economic instability. In one million loose-prior HLT draws, a complex-QZ screen falsely rejects 464,685 draws that real QZ accepts with median residual 8.8×10^{-14}. A Cayley real-part screen matches real QZ on all draws, while doubling accepts 84,606 non-unique cases.

Keywords: DSGE estimation; determinacy; generalized Schur; Bayesian support; GPU computing.

1 Introduction

Determinacy restrictions shape DSGE posteriors. In a Bayesian estimation, the posterior density is proportional to

$$p(\theta \mid Y) \propto L(Y \mid \theta)\pi(\theta)\mathbf{1}\{\theta \in \Theta_D\},$$

where $L(Y \mid \theta)$ is the likelihood, $\pi(\theta)$ is the prior, and Θ_D is the determinacy region. A wrong numerical definition of Θ_D changes the estimand: it truncates the prior, distorts the posterior, and can make two implementations disagree even when their likelihood code is otherwise identical.

This paper proves a small but important point for first-order DSGE estimation: complex roots are not complex economics. For real linear rational-expectations models, complex generalized eigenvalues inside the unit circle represent damped oscillations in a real state space. They should not be rejected. A draw should be rejected when the stable-root count is wrong, when the stable subspace fails the graph condition, or when the recovered policy rule fails the residual check. A complex-valued intermediate Schur basis is not enough.

The result matters for GPU estimation. Generalized QZ is the reference method for linear rational-expectations models (Blanchard and Kahn, 1980; Klein, 2000; Sims, 2002), but full QZ is hard to port efficiently to JAX/GPU. Faster screens such as Cayley transforms and doubling iterations are useful only if they preserve the posterior support. The HLT audit below shows that a real-part Cayley screen can match real QZ on the tested prior support, while doubling is too permissive as a final support classifier.

2 Econometric Object

Let $(D(\theta), E(\theta))$ be the real generalized pencil from the linearized DSGE model at parameter vector θ. Generalized eigenvalues solve

$$\det(D(\theta) - \lambda E(\theta)) = 0.$$

Let n_s be the number of predetermined states. Define the econometric determinacy set as

$$\Theta_D = \{\theta : \#\{|\lambda_i(\theta)| < 1\} = n_s,\ Z_x(\theta) \text{ is nonsingular},\ r(\theta) \le \tau\},$$

where $Z = (Z_x', Z_y')'$ spans the stable deflating subspace, Z_x is the state block, Z_y is the jump-variable block, $r(\theta)$ is the relative residual of the recovered policy rule, and τ is a numerical tolerance. This definition follows the Blanchard-Kahn logic: exactly one stable path must exist for each state.

A numerical support classifier is valid on a parameter region only if it returns the same indicator as $\mathbf{1}\{\theta \in \Theta_D\}$. A classifier that rejects complex intermediate objects rather than failed real-domain solutions is a different econometric model.

Assumption 1 (real regular pencil)**.** *For the parameter vector under consideration, D and E are real, the pencil (D, E) is regular, and no generalized eigenvalue lies on the unit circle.*

Assumption 2 (graph condition)**.** *The stable deflating subspace has dimension n_s. For any full-rank basis $Z = (Z_x', Z_y')'$ of this subspace, the state block Z_x is nonsingular.*

3 Proposition and Proof

Proposition 1 (real policy from a complex stable basis)**.** *Under Assumptions 1 and 2, the first-order policy rule*

$$y_t = Gx_t, \qquad G = Z_y Z_x^{-1},$$

is real. Complex generalized eigenvalues with modulus below one imply damped real cycles, not instability. If a complex-QZ implementation returns a policy matrix with economically meaningful imaginary entries despite the correct root count and graph condition, the failure is numerical extraction, not a rejection of θ from Θ_D.

The intuition is simple. A real model maps real states into real choices. Complex arithmetic is a representation device: one complex stable root and its conjugate describe a two-dimensional real rotation that contracts over time. The real and imaginary parts of the associated eigenvector span the real plane on which that cycle lives.

Proof. Because D and E are real, conjugation preserves generalized eigenpairs. If $Dv = \lambda Ev$, then $D\bar{v} = \bar{\lambda}E\bar{v}$. Hence non-real generalized eigenvalues and eigenvectors occur in conjugate pairs. The set $\{\lambda : |\lambda| < 1\}$ is closed under conjugation, so the stable deflating subspace is closed under conjugation.

Every conjugation-closed complex subspace has a real basis. For each non-real vector $v = a + ib$, replace the conjugate pair $(v, \bar{v})$ by the real vectors $a = \operatorname{Re} v$ and $b = \operatorname{Im} v$. These vectors span the same real invariant plane. Repeating this operation yields a real full-rank stable basis $Z = (Z_x', Z_y')'$.

Assumption 2 makes Z_x nonsingular, so the stable subspace is the graph of $G = Z_y Z_x^{-1}$. Since both Z_y and Z_x are real, G is real. Moreover, if $\tilde{Z} = ZT$ is any other complex basis of the same subspace with invertible T, then

$$\tilde{Z}_y \tilde{Z}_x^{-1} = Z_y T (Z_x T)^{-1} = Z_y Z_x^{-1}.$$

Thus the policy matrix is basis-invariant in exact arithmetic. Non-zero imaginary entries in a computed policy matrix reflect finite precision, ill-conditioning, or a failure to keep conjugate pairs aligned.

Finally, a complex stable root $\lambda = a + ib$ corresponds in real coordinates to a 2×2 block with eigenvalues $a \pm ib$. Powers of this block decay at rate $|\lambda|$. The path oscillates but converges. This is real stability. □

4 Monte Carlo Audit

The proposition is algebraic. The relevant econometric question is whether a wrong complex rejection rule changes the posterior support in a model used for estimation. The audit draws one million loose-prior parameter vectors from the HLT model and classifies each draw using four screens: the old complex-QZ classifier, real QZ, a Cayley real-part classifier, and doubling. The audit file is `benchmarks/results/qz_real_cayley_hlt_loose_1m_stream.json`.

Table 1: Support classification in one million loose-prior HLT draws

Classification event	Count	Share
Real QZ accepts: unique stable solution	702,789	70.3%
Real QZ rejects: indeterminate	256,850	25.7%
Real QZ rejects: no stable solution	40,361	4.0%
Complex-QZ accepts: unique stable solution	238,104	23.8%
Complex-QZ rejects as "complex solution"	464,685	46.5%
Real QZ accepts those "complex" cases	464,685	46.5%
Cayley real-part disagrees with real QZ	0	0.0%
Doubling accepts a real-QZ non-unique case	84,606	8.5%

Notes: The median real-QZ residual among accepted draws is 8.8×10^{-14}; the maximum is 1.0×10^{-11}. The Cayley real-part screen uses the same support decision as real QZ in this audit, but the statement is model-support evidence, not a global theorem.

Table 1 shows a large false-rejection problem. The complex-QZ classifier removes 46.5 percent of the tested support because the extracted policy matrix is complex. Real QZ accepts all of these draws and rejects none of the draws that complex QZ labels unique. The old rule therefore changes Θ_D by excluding many real, stable, unique solutions.

The table also separates speed devices from support classifiers. Doubling finds all real-QZ unique cases, but it accepts 84,606 non-unique cases. It is therefore useful as a candidate solver, warm start, or intermediate screen, but not as the final posterior support test. By contrast, the Cayley real-part screen has zero support disagreements with real QZ in the tested HLT region.

5 Implications for Estimation

The posterior sampler should use the following support hierarchy. First, screen the stable-root count. Second, verify the graph condition. Third, recover a real policy matrix and check the model residual. Complex intermediate objects should trigger realification or fallback to real QZ; they should not trigger economic rejection by themselves.

This hierarchy gives a practical GPU path. JAX can run a Cayley real-part screen and batched residual checks on the GPU. Real QZ remains the reference validator on CPU until a reliable GPU-native generalized QZ is available. Doubling remains useful for speed, but only under a verification layer because it admits indeterminate points in the HLT audit.

6 Conclusion

A complex-QZ "complex solution" label is not a determinacy diagnosis. In a real DSGE model, complex stable roots are real damped cycles. The valid determinacy restriction is root count plus the graph condition plus residual accuracy. Applying this distinction to one million HLT draws changes the accepted support from 238,104 to 702,789 draws relative to the old complex-QZ screen. For estimation, this is not cosmetic: it changes the posterior support.

References

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